



UNIVERSITY OF CENTRAL PUNJAB LAHORE

Assignment 2 CLO Mapping: CLO-2

Course Code	MAT233	Semester	Spring 2026
Course Title	Linear Algebra		
Instructor	Adnan Rafiq Siddiqui		
Assignment Given Date	03-04-2026	Submission Date	09-04-2026
Total Marks	30		

Sr.	Name of student	Registration no.	Section	Assignment marks	Total marks
1					

Question No.1

[10 marks]

Let $(y + 2)^2 = \frac{1}{2}(x - 1)$ be a parabola. **Find** the equation of Parabola after Rotating it by an angle of 90° anticlockwise.

Question No.2

[10 marks]

Each component is represented using basic shapes such as rectangles and circles.

Design Your UI Layout Using Basic Geometric Shapes

Use the coordinate plane to create a basic mobile login screen with the following components:

- **Screen Container (rectangle):** Vertices at $(-1,1)$, $(-1,8)$, $(3,8)$, $(3,1)$
- **Login Button (triangle):** Vertices at $(1,2)$, $(1.5,3)$, $(2,2)$
- **Username Field (rectangle):** Vertices at $(2,5)$, $(2,6)$, $(0,6)$, $(0,5)$

a) **Translation:** Move the layout 2 units right and 1 units down to simulate positioning it on a different screen size or resolution.

b) **Reflection:** Move the layout about the line $y = -x$.

Question No. 3**[10 marks]**

Your mission is to encrypt a classified phrase “**Move the Target**” using Hill Cipher before transmission to headquarters. Your Task

- i. Remove spaces and convert to uppercase, if needed, **pad** the message with an extra letter.
- ii. Convert letters to numbers using the standard alphabet mapping (A = 1, B = 2, ..., Z = 26).
- iii. Use the following **key matrix** K for encryption

$$K = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$$

- iv. **Encrypt** the message using matrix multiplication modulo 26.
- v. Convert the result back to letters and write the final encrypted message.